

The Mean Boundary-Layer Profile

How pyMack computes the steady laminar compressible base flow
whose profiles become the coefficients of the stability equations

Personal reference note

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Abstract

Before any stability question can be asked, the disturbance must have something to ride on: a steady, laminar, compressible boundary layer. This note explains — assuming no prior exposure — how pyMack obtains that base flow. It states the governing compressible Navier–Stokes equations and their non-dimensionalisation, defines the dimensionless groups and the local Mack length that fix the solver’s units, and then reduces the flat-plate mean flow to a small self-similar boundary-value problem for the profiles $\bar{U}(y), \bar{T}(y), \bar{\rho}(y), \bar{\mu}(y)$ and their wall-normal derivatives. Every equation, boundary condition, coefficient, and default is transcribed from the actual code in `pymack/`, cited by file. These profiles — and, crucially, their y -derivatives — are precisely the *known coefficients* that enter the linearised disturbance equations, which are derived in the companion note.

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How to read this. Boxes marked “**What problem is solved here**” state, at each stage, exactly which equation and which boundary conditions are involved. *In plain words* notes give the intuition; grey Remark asides hold secondary or fallback detail you can skip on a first read.

Notation. Overbars are mean (base-flow) quantities. A prime on a mean quantity is $D \equiv d/dy$ (or $d/d\eta$ in the similarity coordinate, as noted in context). The rows below cover only the base-flow, transport, and scale symbols this note uses; the disturbance/eigenvalue symbols live in the companion notes.

Base flow & transport

$\bar{U}, \bar{T}, \bar{\rho}=1/\bar{T}$	mean velocity, temperature, density
$\bar{\mu}, \bar{\kappa}$	viscosity, conductivity
$C=\bar{\mu}/\bar{T}, K, B$	transport groups, §2

Scales & parameters

M_e, Re, Pr, γ	Mach, Reynolds, Prandtl, c_p/c_v
R_g, a_e	specific gas constant, edge sound speed ($a_e^2=\gamma R_g T_e$)
$L^*=\sqrt{\nu_e x/U_e}$	Mack length, Eq. (7)
$R=\sqrt{Re_x}, F$	Mack Reynolds, reduced frequency, Eq. (8)
$\alpha, \alpha_L=\alpha L^*, \delta^*, \theta$	streamwise wavenumber, scaled wavenumber, displacement/momentum thickness

The disturbance rides on a steady laminar flow, so we need that flow — and its y -derivatives — before anything else. This note computes it.

1 Governing equations and non-dimensionalisation

The flow obeys the compressible Navier–Stokes equations — conservation of mass, momentum, and energy for a viscous, heat-conducting perfect gas:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0, \quad (\text{mass}) \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau}, \quad (\text{momentum}) \quad (2)$$

$$\rho c_p \frac{DT}{Dt} = \frac{Dp}{Dt} + \nabla \cdot (\kappa \nabla T) + \Phi, \quad (\text{energy}) \quad (3)$$

$$p = \rho R_g T, \quad (\text{state}) \quad (4)$$

where $\mathbf{u} = (u, v)$ is velocity, ρ density, T temperature, p pressure, μ viscosity, κ conductivity, c_p specific heat, R_g the specific gas constant (plain R is reserved for the Mack Reynolds number, Eq. (8)), and $D/Dt = \partial_t + \mathbf{u} \cdot \nabla$ the material derivative. The two remaining symbols are the Newtonian **viscous stress tensor** and the **dissipation function**,

$$\boldsymbol{\tau} = \mu \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right), \quad \Phi = \boldsymbol{\tau} : \nabla \mathbf{u}. \quad (5)$$

We will not need their closed form again: all that matters downstream is that they are linear in velocity gradients. These four equations are applied *twice* — once to obtain the steady mean flow (this note), once linearised for the disturbance (the companion note).

Non-dimensionalisation and default gas constants. Velocities are scaled by the edge speed U_e , temperature by T_e , density by ρ_e , viscosity by μ_e , pressure by $\rho_e U_e^2$. Because the gas is perfect and pressure is nearly uniform across the layer, $\bar{p} = 1/\bar{T}$ in these units. Three dimensionless groups control everything,

$$M_e = \frac{U_e}{a_e}, \quad a_e = \sqrt{\gamma R_g T_e}, \quad Re = \frac{\rho_e U_e L_{\text{ref}}}{\mu_e}, \quad Pr = \frac{\mu_e c_p}{\kappa_e}, \quad (6)$$

where a_e is the edge speed of sound and R_g the specific gas constant. with default air values $\gamma = 1.4$, $Pr = 0.72$, Sutherland constant $S = 110.4$ K, and power-law exponent $\omega_\mu \approx 0.7$ when the power-law viscosity is used.

The Mack length and Reynolds scales. Boundary layers grow like $\sqrt{x}$, so the natural local scale at station x is

$$L^* = \sqrt{\frac{\nu_e x}{U_e}}, \quad \nu_e = \frac{\mu_e}{\rho_e}. \quad (7)$$

With $L_{\text{ref}} = L^*$ the Reynolds number becomes the square root of the usual one, and a fixed physical frequency f maps to a reduced frequency F (both consumed later, by the neutral-curve and N -factor computations in the companion notes):

$$R = \frac{U_e L^*}{\nu_e} = \sqrt{Re_x}, \quad Re_x = \frac{U_e x}{\nu_e}; \quad F = \frac{2\pi f \nu_e}{U_e^2}; \quad \alpha_L = \alpha L^*, \quad (8)$$

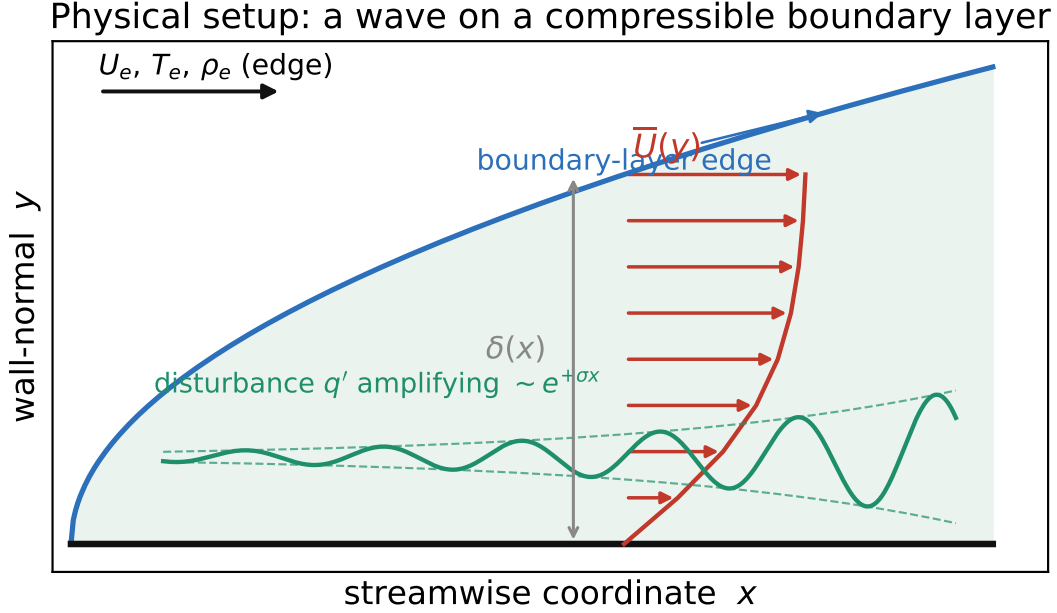


Figure 1: The physical setup. A steady compressible boundary layer grows along a wall; its mean velocity $\bar{U}(y)$ and temperature $\bar{T}(y)$ vary across the layer of thickness $\delta(x)$. Stability theory tracks one small disturbance $q'(x, y, t)$ riding in this layer and asks whether it amplifies downstream — but first the mean profiles $\bar{U}, \bar{T}$ must be computed, which is the subject of this note.

where α is the streamwise wavenumber. R is the horizontal axis of every neutral curve and α_L the vertical axis. Conversions live in `pymack/scales.py`; the solver itself is dimensionless. A profile may be stored on a y/δ^* or a y/L^* grid; the constant δ^*/L^* converts between them, with $D \mapsto (\delta^*/L^*)^{-1}D$.

2 The base flow

What problem is solved here. Solve a *boundary-value problem* (BVP) — not an eigenvalue problem — for the steady laminar profiles $\bar{U}(y), \bar{T}(y), \bar{p}(y), \bar{\mu}(y)$ and their y -derivatives. A BVP fixes the data at both ends ($y = 0$ and $y \rightarrow \infty$) and has a single solution; an eigenvalue problem instead leaves a parameter free and asks for the special values that admit a non-zero solution. These profiles become the known coefficients of the disturbance equations.

For a flat plate the mean flow is *self-similar*: profiles at different x collapse onto one curve in a stretched coordinate η . With the dimensionless stream function $f(\eta)$, $f' = \bar{U} = u/U_e$ (a prime is $d/d\eta$), momentum and energy reduce to two coupled ODEs,

$$(C f'')' + f f'' = 0, \quad (9)$$

$$(B \bar{T}')' + f \bar{T}' + (\gamma - 1) M_e^2 C (f'')^2 = 0, \quad (10)$$

with transport groups

$$C = \frac{\mu/\mu_e}{T/T_e} \text{ (Chapman-Rubeson),} \quad K = \frac{\kappa}{c_p \mu_e}, \quad B = \frac{K}{\bar{T}} \left(= \frac{C}{Pr} \text{ at constant } Pr \right). \quad (11)$$

The last term in (10) is aerodynamic heating — it makes a hypersonic layer hot near the wall and couples energy to velocity. The conductivity model is $\kappa = \mu c_p / Pr$ (so $\bar{\kappa} = \bar{\mu} / Pr$ at constant Pr); the transport derivatives $d\mu/dT$ etc. are taken analytically from the chosen viscosity law.

Wall and edge conditions.

$$\underbrace{f(0) = 0, f'(0) = 0}_{\text{no slip}}, \quad \underbrace{f'(\infty) = 1, \bar{T}(\infty) = 1}_{\text{match the edge}}, \quad \begin{cases} \bar{T}'(0) = 0 & \text{(adiabatic wall)} \\ \bar{T}(0) = T_w/T_e & \text{(isothermal wall)} \end{cases} \quad (12)$$

For an adiabatic wall the recovered temperature is the standard result (stated, not derived) $T_{aw}/T_e = 1 + \frac{1}{2}(\gamma-1)\sqrt{Pr} M_e^2$. (For a cone, the edge state itself comes first, from the Taylor–Maccoll equation behind the conical shock; `taylor_maccoll_edge_state.py`.)

Solution and the coordinate transform. Equations (9)–(10) are solved as a 5-state first-order system $[f, \bar{U}, \bar{U}', \bar{T}, \bar{T}']$ by `scipy.integrate.solve_bvp` (collocation + Newton; `pymack/baseflow.py`), integrated to $\eta_{\max} \approx 40$ (well beyond the edge). The similarity coordinate maps to the physical one by

$$\frac{y}{L^*} = \sqrt{2} \int_0^\eta \frac{\bar{T}}{T_e} d\eta', \quad \frac{\delta^*}{L^*} = \sqrt{2} \int_0^\infty (\bar{T} - \bar{U}) d\eta, \quad \frac{\theta}{L^*} = \sqrt{2} \int_0^\infty \bar{U}(1 - \bar{U}) d\eta, \quad (13)$$

where the displacement thickness δ^* measures how far the wall effectively pushes the outer flow outward and the momentum thickness θ the momentum deficit. The solver returns, on any grid, $\bar{U}, \bar{U}', \bar{U}'', \bar{T}, \bar{T}', \bar{T}'', \bar{\rho}, \bar{\mu}, d\mu/dT, d^2\mu/dT^2, \bar{\kappa}, d\kappa/dT, d^2\kappa/dT^2$.

Two base-flow implementations. `baseflow.py` contains two equivalent self-similar formulations sharing the 5-ODE structure above. Equations (9)–(10) and the $\sqrt{2}$ transform (13) are the `CompressibleBlasiusProfile` form. The temporal solver and the worked example (in the companion Numerical Methods note) use `make_flatplate_profile` (`FlatPlateProfile`; Ozgen & Kircali transport laws), an alternative normalisation ($f' = \bar{U}/\bar{T}$, $\frac{1}{2}$ factors, $y/L^* = \eta$, no $\sqrt{2}$). The two give the same physical profiles; only the similarity bookkeeping differs.

These profiles $\bar{U}(y), \bar{T}(y), \bar{\rho}(y), \bar{\mu}(y)$ and their y -derivatives are exactly the coefficients that appear in the linearised disturbance equations; that linearisation and the resulting ordinary differential equations are derived in the companion “Derivation of the Disturbance Equations” note.

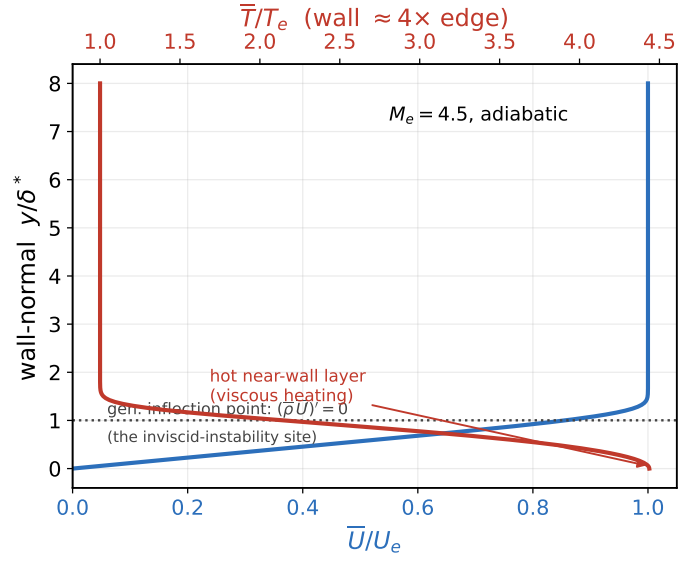


Figure 2: A computed compressible base flow ($M_e = 4.5$, adiabatic). $\bar{U}(y)/U_e$ rises monotonically to the edge; $\bar{T}(y)$ bulges near the wall from viscous heating. These curves and their y -derivatives are the known coefficients $\bar{U}, \bar{T}, \dots$ in the disturbance equations. The dotted line marks the generalised inflection point, the seat of the inviscid instability (discussed in the companion notes on the instability mechanisms).